

Harihara Yashwanth Veldanda

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AI/ML Engineer shipping end-to-end: model fine-tuning, agentic pipelines, and cloud deployment - MS Data Science, ASU.

TECHNICAL SKILLS

Languages: Python, SQL, R, C, C++, HTML, CSS, JavaScript

Frameworks & Libraries: Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch, Keras, Matplotlib, Node.js, React

Databases & Cloud: Google BigQuery, Azure, AWS, MySQL, PostgreSQL

Tools: VS Code, Jupyter, Power BI, Tableau, Apache Spark, Docker, Git

MLOps & Deployment: Docker, Azure ML, GitHub Actions, LangGraph, MLflow, CI/CD

PROFESSIONAL EXPERIENCE

MathCo

Bengaluru, India

Data Analyst (Intern Analyst → Data Analyst)

Jan 2023 - Jul 2024

- Partnered with Customers, Insights and Strategies team for **WALMART** (client) and executed incremental analysis, delivered insights that drove a 10% increase in overall customer spending and acquisition of 30% new Walmart+ members.
- Built an NLP pipeline to surface churn drivers from sentiment data, powering fixes that cut cancellations by 20%.
- Applied **KNN** for customer audience segmentation, enabling targeted strategies that led to a 30% increase in customer engagement.
- Built a gradient-boosted retention model using **Python, BigQuery** on engagement timing and behavioral signals, lifting member retention by 40%.
- Designed and evaluated 15+ **A/B tests (hypothesis testing, statistical significance)** on offers and promotions, driving a 15% increase in conversion rates.
- Delivered stakeholder-facing dashboards translating model outputs into business recommendations.
- Designed predictive models for luxury fashion, identifying 90% of key sales drivers, boosting sales by 10%.
- Constructed a simulator for stock estimation (MAPE 12%) using **Power BI** and internal tools.
- Tech Stack:** Python, SQL, Google BigQuery, PySpark

PERSONAL PROJECTS

Applied AI Engineering – Tyche Labs| LekhaAI

May 2025 - Present

- Fine-tuned a **FLUX** diffusion model using **LoRA** on ~1,500 image-prompt pairs to improve layout quality and text rendering in generated invitations.
- Ran iterative experiments on prompt structure, resolution bucketing, and guidance scaling, improving visual fidelity by ~25% and reducing text artifacts by ~30% compared to the base model.
- Deployed containerized GPU inference via **Azure ML** Online Endpoints (A100), optimizing cold-start latency and model loading.
- Integrated **GPT-4.0** nano to structure user prompts and handle conversational refinement before image generation.
- Developed the full-stack pipeline (React + TypeScript frontend, Azure Functions backend) to support real-time generation and template preview.
- Deployed the application to production with CI/CD (GitHub + Vercel) and custom domain configuration.
- Demo: [Fine-tuned model](#), [Lekha-AI](#), [Video-demo](#)
- Tech Stack:** Python, PyTorch, Diffusers, Azure ML, Docker, REST APIs, React, TypeScript

Multi-Agent SQL/PySpark Query Optimizer – SQLSparkOptimizer

Jun 2026 - Present

- Shipped a pip-installable package (CLI + library) with a live demo; 6-agent LangGraph pipeline auto-rewrites SQL→PySpark.
- Proved every rewrite output-identical via DuckDB/Spark re-execution, closing the validation gap AQE/Snowflake leave open.
- Validated unmodified on a live Databricks serverless cluster (1.54x real speedup); stress-tested 1,000 queries in 2.6 min.
- Achieved 2.22x avg speedup (up to 4.3x on joins) at \$0 cost, with LLM escalation for novel anti-patterns no rule covers.
- Tech Stack:** Python, PySpark, DuckDB, SQLGlot, LangGraph, Neo4j, pgvector, MLflow

ACADEMIC PROJECTS

Retrieval-Augmented Generation (RAG) Assistant for Workday HR Workflows

Jan 2025 - May 2025

- Built a containerized RAG assistant indexing 10K+ HR records via RocksDB and FAISS for secure, low-latency retrieval.
- Boosted query accuracy by 83% using context-grounded semantic search (cosine similarity < 0.5).
- Automated common HR queries and personalized communications, reducing manual support workload by an estimated 40%.
- Tech Stack:** Python, SQL, RocksDB, Docker, FastAPI

Construction of Medical Chatbot using Knowledge Graph in Neo4j

Aug 2022 - Dec 2022

- Designed a Neo4j knowledge graph linking diseases, symptoms, and treatments for intelligent query handling.
- Integrated NLP-based entity extraction and intent detection, improving query accuracy by 25%.
- Tech Stack:** Python, Neo4j, Streamlit, Docker, RestAPI, NLTK, Scikit-learn

EDUCATION

Arizona State University

Aug 2024 - May 2026

- Master of Science - Data Science and Analytics Engineering
- Relevant Coursework: Artificial Intelligence, Timeseries Forecasting, Statistical Machine Learning, Data Mining

GPA : 3.87